



Vel Tech
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AI-3D ULPIN Generation and vertical Property Mapping System

TEAM: Terra Mind

Problem Statement :

- **Traditional 2D land records** cannot accurately represent vertical and underground properties.
- Modern properties include **multi-storey apartments, basements, parking areas, tunnels, and utility networks**.
- 2D systems create difficulties in **uniquely identifying properties and ownership rights**.
- They also make it difficult to **define, update, and manage property boundaries**.
- These limitations can lead to **ownership conflicts and property disputes**.
- There is a need for an **AI-powered 3D cadastral system** for modern urban environments.
- The system can integrate **GIS, LiDAR, drone imagery, GNSS/CORS, DEM/DSM, and building plans**.
- It can **automatically map vertical properties, generate 3D ULPINs, validate boundaries, and provide 3D visualization** for efficient land governance and infrastructure planning

Abstract :

- **3D ULPIN Generation and Vertical Property Mapping System** is an advanced geospatial framework for modern land administration.
- Addresses the limitations of **conventional 2D cadastral systems** in rapidly urbanizing areas.
- Creates **unique 3D spatial identities** for Surface land parcels ,Multi-storey apartments ,Underground infrastructure ,Parking areas ,Vertically distributed property rights
- Integrates multiple geospatial datasets like Drone imagery ,LiDAR and 3D point clouds ,GIS parcel layers ,Building floor plans ,GNSS/CORS coordinates ,DEM/DSM datasets .
- Develops an **accurate and interoperable 3D cadastral database**.
- Uses **AI/ML techniques** for Automated building extraction ,Floor-level segmentation ,Vertical parcel delineation and Intelligent topology validation.
- Reduces **manual effort** and improves **spatial accuracy**.
- Introduces a **standardized 3D ULPIN generation mechanism** for uniquely identifying individual volumetric property units and ownership rights.
- Supports **3D cadastre and vertical property governance**.
- Enables authorities to manage complex ownership relationships **above and below ground**.
- Helps to reduce Property identification issues ,Ownership disputes ,Property ambiguities.
- Supports **infrastructure and utility planning**.
- Enables more **efficient urban land governance**.
- Provides a **scalable and interoperable** cadastral framework.
- Combines **geospatial technologies with AI/ML** to create an intelligent, future-ready land management system.
- Particularly useful for managing the increasing **vertical and underground dimensions of modern cities**

Description:

- **3D ULPIN System** provides a modern framework for **3D land and property management**.
- Overcomes the limitations of **traditional 2D cadastral systems** in urban areas.
- Creates **unique 3D identities** for land parcels, apartments, parking spaces, and underground infrastructure.
- Integrates **drone imagery, LiDAR, 3D point clouds, GIS, GNSS/CORS, DEM/DSM, and building floor plans**.
- Uses **AI/ML** for automated building extraction, floor segmentation, parcel delineation, and topology validation.
- Generates **standardized 3D ULPINs** to uniquely identify volumetric property units and ownership rights.
- Helps reduce **property disputes and ambiguities** while supporting infrastructure and utility planning.
- Provides a **scalable, interoperable, and future-ready 3D cadastral system** for efficient urban land governance.
- <https://videotoururl.com/videos/1787681637840-de4cbaba-db53-4be1-b68f-be0a70bbf59c.mp4>
- <https://videotoururl.com/videos/1787681690136-83a6fad5-3520-4a7e-ac63-22b8b925fee9.mp4>

Building the Future of Urban Management



Unique 3D Identities

Unique 3D identities for surface, vertical, and underground properties.



Reduced Ownership Disputes

Clearly defines property boundaries and ownership rights.



Better Urban Planning

Supports accurate planning of urban infrastructure.



Efficient Land Governance

Manages and updates property records digitally.



Improved Infrastructure Management

Maps and monitors underground utilities and infrastructure.



Higher Accuracy

Combines technologies for precise property mapping.

Future-Ready
Smart Cities

Feasibility of 3D ULPIN Generation and Vertical Property Mapping System

Technical Feasibility

GIS, LiDAR, drone imagery, GNSS/CORS, DEM/DSM, and 3D visualization technologies are already available and can be integrated.

AI/ML Feasibility

Existing machine-learning models can be adapted for building extraction, floor segmentation, property boundary detection, and topology validation.

Data Feasibility

Sample GIS, building, LiDAR, and elevation datasets can be used to develop and test the prototype.

Implementation Feasibility

A web-based system can be developed using technologies such as Python, FastAPI, PostgreSQL/PostGIS, React, and CesiumJS/Three.js.

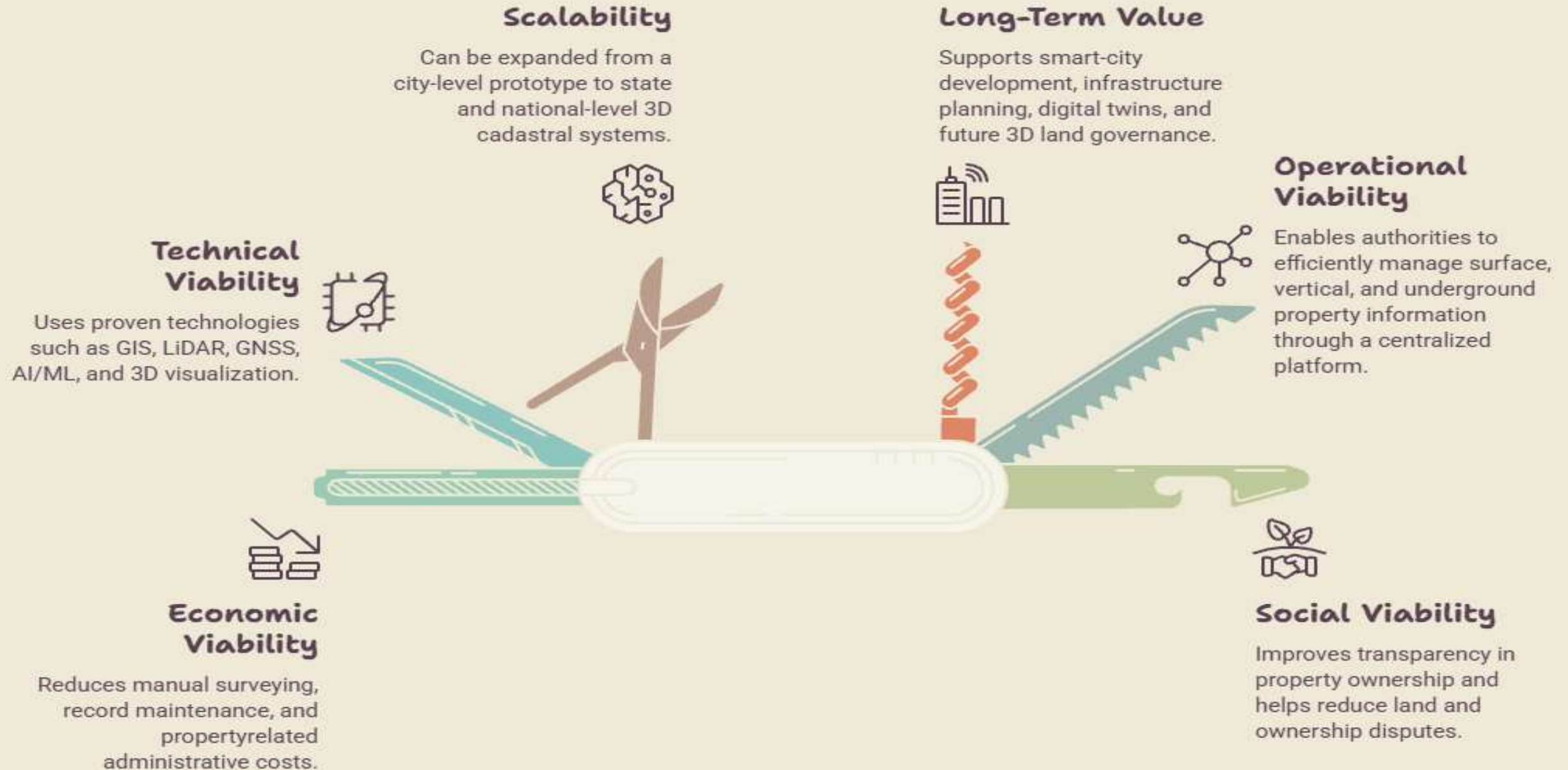
Scalability

The prototype can later be expanded to support larger cities, government land records, and smart-city applications.

Hackathon Feasibility

A focused MVP can demonstrate 3D mapping, property segmentation, ULPIN generation, and interactive visualization within the available time.

Viability of the 3D ULPIN Generation and Vertical Property Mapping System



THANK YOU